MATH 6660 — Homework 2

Hasan

Problem 1

Problem. For the periodic boundary value problem

$$\textbf{ODE}: \ -u'' + q(x)\,u = f(x), \qquad \textbf{BC1}: \ u(-\pi) = u(\pi), \qquad \textbf{BC2}: \ u'(-\pi) = u'(\pi), \tag{1}$$

with $q > 0$ for all $x \in [-\pi, \pi]$, derive a Galerkin orthogonality expression and deduce from it an appropriate *energy norm*.

Solution

Step 1: variational form. Multiply (1) by a test function v and integrate over $(-\pi, \pi)$:

$$\int_{-\pi}^{\pi} \big(-u'' + q\,u\big) v \,\mathrm{d}x = \int_{-\pi}^{\pi} f\, v \,\mathrm{d}x. \tag{2}$$

Integrate the second-derivative term by parts once:

$$\int_{-\pi}^{\pi} \big(u'v' + q\,u\,v\big)\,\mathrm{d}x \ - \ \underbrace{\big[\,u'v\,\big]_{-\pi}^{\pi}}_{\text{boundary term}} \ = \ \int_{-\pi}^{\pi} f\, v \,\mathrm{d}x. \tag{3}$$

The boundary term is

$$\big[\,u'v\,\big]_{-\pi}^{\pi} = u'(\pi)\,v(\pi) - u'(-\pi)\,v(-\pi) = u'(\pi)\big[\,v(\pi) - v(-\pi)\,\big] = 0, \tag{4}$$

where **BC2** was used to replace $u'(-\pi)$ by $u'(\pi)$, and the last equality holds provided v is periodic. That requirement is what fixes the space:

$$V = \Big\{ v : \ \int_{-\pi}^{\pi} \big(v^2 + (v')^2\big)\,\mathrm{d}x < \infty, \quad v(-\pi) = v(\pi) \Big\}. \tag{5}$$

Dropping (4) from (3) gives the variational problem: find $u \in V$ such that

$$\int_{-\pi}^{\pi} \big(u'v' + q\,u\,v\big)\,\mathrm{d}x = \int_{-\pi}^{\pi} f\, v \,\mathrm{d}x \qquad \forall\, v \in V. \tag{6}$$

Note. Only **BC1** is built into V in (5); **BC2** is *natural*, never imposed on the space. Two points are worth being precise about. First, BC2 *alone* does not make (4) vanish: it lets the two endpoint terms be collected into the single factor $u'(\pi)\big[v(\pi) - v(-\pi)\big]$, which then vanishes because $v \in V$ is periodic. It takes both. Second, BC2 is genuinely *recovered* from (6), in two steps. First test with v compactly supported in $(-\pi, \pi)$: this gives $u'' = qu - f$ distributionally, so $u'' \in L^2$ and $u \in H^2$ — which is what makes $u'(\pm\pi)$ meaningful and the next step legal. Now reverse the integration by parts for general $v \in V$: $\int_{-\pi}^{\pi} \big(-u'' + qu - f\big) v \,\mathrm{d}x + \big[u'v\big]_{-\pi}^{\pi} = 0$. The integral vanishes by what was just shown, leaving $\big(u'(\pi) - u'(-\pi)\big)v(\pi) = 0$; choosing any $v \in V$ with $v(\pi) \neq 0$ forces $u'(-\pi) = u'(\pi)$. Trial and test space coincide.

Step 2: the finite element method. Choose any finite-dimensional $V_h \subset V$. The FEM is: find $U \in V_h$ such that

$$\int_{-\pi}^{\pi} \big(U'v' + q\,U\,v\big)\,\mathrm{d}x = \int_{-\pi}^{\pi} f\,v\,\mathrm{d}x \qquad \forall\, v \in V_h. \tag{7}$$

Step 3: subtract. Because $V_h \subset V$, the exact solution u satisfies (6) in particular for every $v \in V_h$:

$$\int_{-\pi}^{\pi} \big(u'v' + q\,u\,v\big)\,\mathrm{d}x = \int_{-\pi}^{\pi} f\,v\,\mathrm{d}x \qquad \forall\, v \in V_h. \tag{8}$$

Subtracting (7) from (8), the right-hand sides cancel. With the error $e := u - U$,

$$\boxed{\int_{-\pi}^{\pi} \big(e'v' + q\,e\,v\big)\,\mathrm{d}x = 0 \qquad \forall\, v \in V_h} \tag{9}$$

which is the Galerkin orthogonality expression for this problem.

Step 4: read off the energy norm. Define the bilinear form appearing on the left of (6):

$$a(v,w) := \int_{-\pi}^{\pi} \big(v'w' + q\,v\,w\big)\,\mathrm{d}x. \tag{10}$$

Then (9) reads

$$a(e,v) = 0 \qquad \forall\, v \in V_h, \tag{11}$$

i.e. the error is orthogonal to the whole subspace *with respect to* a. For that statement to be genuine orthogonality, a must be an inner product on V:

- *bilinear* — each slot is linear, by inspection of (10);
- *symmetric* — $a(v,w) = a(w,v)$, again by inspection;
- *positive definite* —

$$a(v,v) = \int_{-\pi}^{\pi} \Big[\,(v')^2 + q\,v^2\,\Big]\,\mathrm{d}x \;\geq\; 0, \tag{12}$$

  and both terms are non-negative, so $a(v,v) = 0$ forces $v' \equiv 0$ *and*, since $q > 0$, $v \equiv 0$.

Hence a induces a norm, and it is the appropriate energy norm here:

$$\boxed{\|v\|_E := \sqrt{a(v,v)} = \left(\int_{-\pi}^{\pi} \Big[\,(v')^2 + q\,v^2\,\Big]\,\mathrm{d}x\right)^{1/2}} \tag{13}$$

Where $q > 0$ is used. Not decoration. Under periodic conditions, $v' \equiv 0$ only gives $v = \text{const}$, and a constant *is* an element of V in (5) — it is periodic. So $\int (v')^2\,\mathrm{d}x$ alone is a seminorm, not a norm: it cannot see constants. The term $q\,v^2$ in (12) is exactly what rules them out.

Note. Contrast HW1's Dirichlet problem, where $v(0) = 0$ removes the constant for free and $\int a(x)\,(v')^2\,\mathrm{d}x$ was already a norm on its own.

Note. **Standing hypothesis on q.** The problem statement gives only *pointwise* positivity, $q(x) > 0$, and that is all (12) needs: both integrands are non-negative and $q > 0$ forces $v \equiv 0$. Problems 3 and 4, however, use $q_{\max} = \max q$ and $q_{\min} = \min q > 0$, and a uniform positive lower bound is strictly stronger than pointwise positivity. From here on assume in addition that q is continuous on the compact interval $[-\pi, \pi]$; then both extrema are attained and $0 < q_{\min} \leq q(x) \leq q_{\max} < \infty$. The data of Problem 5 satisfy this, with $q_{\min} = 2 - \sqrt{2} \approx 0.586$ at $x = \pi/4$ and $q_{\max} \approx 4.25$.

Problem 2

Problem. Develop a CG(1) method for (1), clearly stating the appropriate spaces, trial, test, and basis functions.

Solution

Mesh. Take the uniform mesh of the hint:

$$h = \frac{2\pi}{M+1}, \qquad x_j = -\pi + j\,h, \quad j = 0, 1, \ldots, M+1, \qquad \text{so } x_0 = -\pi, \quad x_{M+1} = \pi. \tag{14}$$

The elements are $I_j = [x_{j-1}, x_j]$, $j = 1, \ldots, M+1$ — so there are $M+1$ elements. Periodicity identifies $x_0 \equiv x_{M+1}$, leaving $M+1$ *distinct* nodes rather than $M+2$.

Spaces. The continuous space is V from (5). The discrete space is

$$V_h = \big\{\, v \in V : \; v|_{I_j} \in P_1(I_j), \quad j = 1, \ldots, M+1 \,\big\}, \tag{15}$$

continuous piecewise linear functions that are also periodic. By construction $V_h \subset V$, so the method is conforming and Galerkin orthogonality (9) applies verbatim.

Note. $\dim V_h = M+1$. A continuous piecewise linear function on $M+1$ elements carries $M+2$ nodal values $\xi_0, \ldots, \xi_{M+1}$, and periodicity forces $\xi_0 = \xi_{M+1}$, removing one. Compare the Dirichlet problem, where *two* values are removed by prescribing them, leaving M. Here one value is removed by an *identification*, not by prescription.

Basis functions. For the interior nodes $j = 1, \ldots, M$, the usual hat:

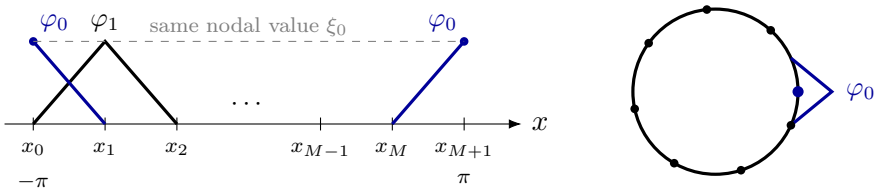
$$\varphi_j(x) = \begin{cases} \dfrac{x - x_{j-1}}{h}, & x_{j-1} \le x \le x_j, \\[2ex] \dfrac{x_{j+1} - x}{h}, & x_j \le x \le x_{j+1}, \\[2ex] 0, & \text{otherwise.} \end{cases} \tag{16}$$

At the glued node there is one *wrap-around* hat:

$$\varphi_0(x) = \begin{cases} \dfrac{x_1 - x}{h}, & x_0 \le x \le x_1, \\[2ex] \dfrac{x - x_M}{h}, & x_M \le x \le x_{M+1}, \\[2ex] 0, & \text{otherwise.} \end{cases} \tag{17}$$

On the interval φ_0 looks like two half-hats, one at each end; on the circle it is a single ordinary hat that the interval happens to cut in two. The nodal property still holds, $\varphi_i(x_j) = \delta_{ij}$, reading x_{M+1} as x_0. Hence

$$V_h = \operatorname{span}\{\varphi_0, \varphi_1, \ldots, \varphi_M\}, \qquad \dim V_h = M+1. \tag{18}$$



on the interval, φ_0 is *two half-hats*

on the circle it is one ordinary hat, at the glued node $x_0 \equiv x_{M+1}$

The two ends of the interval are the same point of the domain, so the two half-hats in (17) are two pieces of one basis function and share a single unknown ξ_0. Cutting the circle open to lay it flat is what splits that hat in half.

Trial and test functions. Trial and test space coincide (Galerkin):

$$\text{trial:} \quad U(x) = \sum_{j=0}^{M} \xi_j \, \varphi_j(x), \quad \xi_j = U(x_j); \qquad\qquad \text{test:} \quad v = \varphi_i, \quad i = 0, 1, \ldots, M. \tag{19}$$

The CG(1) method. Find $U \in V_h$ such that

$$\int_{-\pi}^{\pi} \big(U' v' + q\, U\, v\big)\, \mathrm{d}x = \int_{-\pi}^{\pi} f\, v \, \mathrm{d}x \qquad \forall\, v \in V_h. \tag{20}$$

Substituting (19) and taking $v = \varphi_i$ for $i = 0, \ldots, M$ turns (20) into

$$\sum_{j=0}^{M} \xi_j \int_{-\pi}^{\pi} \big(\varphi_i' \varphi_j' + q\, \varphi_i \varphi_j\big)\, \mathrm{d}x = \int_{-\pi}^{\pi} f\, \varphi_i \, \mathrm{d}x, \qquad \text{i.e.} \qquad \boldsymbol{A}\,\boldsymbol{\xi} = \boldsymbol{b}, \tag{21}$$

with

$$A_{ij} = \int_{-\pi}^{\pi} \big(\varphi_i' \varphi_j' + q\, \varphi_i \varphi_j\big)\, \mathrm{d}x, \qquad b_i = \int_{-\pi}^{\pi} f\, \varphi_i \, \mathrm{d}x, \qquad i, j = 0, 1, \ldots, M. \tag{22}$$

Structure of $\boldsymbol{A}$. On the uniform mesh $\varphi_j' = \pm 1/h$ on each of its two supporting elements, so the stiffness part of (22) is

$$\int_{-\pi}^{\pi} \big(\varphi_i'\big)^2 \, \mathrm{d}x = \frac{2}{h}, \qquad \int_{-\pi}^{\pi} \varphi_i' \varphi_{i\pm1}' \, \mathrm{d}x = -\frac{1}{h}, \tag{23}$$

which is $\frac{1}{h}\,\mathrm{tridiag}(-1, 2, -1)$ — *plus two corner entries*. The functions φ_0 and φ_M share the last element $[x_M, x_{M+1}]$, where $\varphi_0' = +1/h$ and $\varphi_M' = -1/h$, so that element contributes $-1/h$ to $A_{0,M}$. The mass part contributes $\int q\, \varphi_0 \varphi_M \, \mathrm{d}x > 0$ over the same element, of size $O(h)$ — the *opposite* sign, so the two compete rather than reinforce. For h small the $-1/h$ term dominates and

$$A_{0,M} = A_{M,0} \neq 0,$$

which holds for every M in this assignment.

Note. On a very coarse mesh the two contributions can cancel: the assembled entry is $+0.047$ at $M = 4$ and -0.234 at $M = 5$, so it passes through zero in between. Nothing in the method depends on the *value* of that entry — it is the *sparsity pattern* that distinguishes the periodic problem, and the pattern is unaffected.

So $\boldsymbol{A}$ is $(M+1) \times (M+1)$, tridiagonal *with wrap-around corners* — a circulant sparsity pattern, not a tridiagonal one.

$\boldsymbol{A}$ is symmetric because $a(\cdot,\cdot)$ in (10) is, and positive definite because (12) gives $a(v,v) > 0$ for $v \not\equiv 0$. That is exactly the hypothesis the conjugate gradient solver needs.

Note. $q > 0$ is what makes $\boldsymbol{A}$ nonsingular. With $q \equiv 0$ the constant vector $(1, 1, \ldots, 1)$ would lie in its null space — the discrete echo of the seminorm remark at the end of Problem 1.

Note. Two unrelated things both abbreviate to "CG": **CG(1)** is *continuous Galerkin* of degree 1, the method; `cg` in `scipy.sparse.linalg` is *conjugate gradient*, the linear solver.

Problem 3

Problem. Show that the CG(1) solution (20) is optimal in the energy norm (13), and use this to prove an *a priori* error estimate in the energy norm.

Solution

Part 1: optimality. Let $v \in V_h$ be arbitrary. Claim:

$$\boxed{\|u - U\|_E \;\le\; \|u - v\|_E \qquad \forall\, v \in V_h} \tag{24}$$

Write the quantity on the right in terms of the error $e = u - U$:

$$u - v = \underbrace{(u - U)}_{= e} + \underbrace{(U - v)}_{=:\, w \in V_h}, \tag{25}$$

where $w = U - v \in V_h$ because V_h is a vector space and both U and v lie in it. Expand in the a-inner product (10):

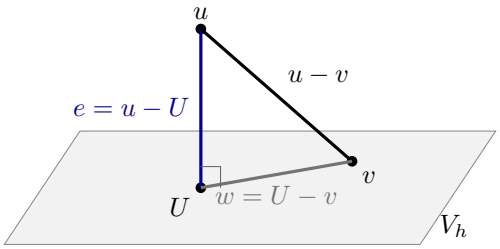
$$\|u - v\|_E^2 = a(e + w,\, e + w) = a(e, e) + 2\, a(e, w) + a(w, w). \tag{26}$$

The middle term vanishes: $w \in V_h$, so Galerkin orthogonality (11) gives $a(e, w) = 0$. Hence

$$\|u - v\|_E^2 = \|e\|_E^2 + \|w\|_E^2 \;\ge\; \|e\|_E^2, \tag{27}$$

and taking square roots gives (24). Equality holds only when $\|w\|_E = 0$, i.e. $v = U$.

Note. (27) is Pythagoras in the a-inner product: U is the a-orthogonal projection of u onto V_h, and e is the perpendicular dropped from u to the subspace. This is Céa's lemma; because the problem is symmetric the constant is exactly 1, so "quasi-optimal" is here genuinely *optimal* — no other member of V_h is closer to u in the energy norm.



Galerkin orthogonality says the blue leg is perpendicular to the whole subspace. Every $v \in V_h$ is the far end of a right triangle whose legs are e and w, so the hypotenuse $\|u - v\|_E$ can never be shorter than $\|e\|_E$ — which is (27).

Part 2: the a priori estimate. (24) holds for *every* $v \in V_h$, so we are free to choose the most convenient one. Take the nodal interpolant $\pi_h u \in V_h$,

$$\pi_h u(x) = \sum_{j=0}^{M} u(x_j)\, \varphi_j(x).$$

Note. $\pi_h u$ really is in V_h: it is continuous piecewise linear by construction, and it is periodic because $\pi_h u(-\pi) = u(-\pi) = u(\pi) = \pi_h u(\pi)$. The single nodal value ξ_0 at the glued node is assigned $u(x_0)$, consistently at both ends.

Assume u is regular enough that $u'' \in L^2(-\pi,\pi)$. The standard interpolation estimates for a continuous piecewise linear interpolant are

$$\|u - \pi_h u\|_{L^2} \le C_2 \, \|h^2 u''\|_{L^2}, \qquad \|(u - \pi_h u)'\|_{L^2} \le C_1 \, \|h \, u''\|_{L^2}. \tag{28}$$

Insert them into the energy norm (13) of the interpolation error, taking h constant and writing $q_{\max} = \max_{[-\pi,\pi]} q$:

$$\|u - \pi_h u\|_E^2 = \underbrace{\|(u - \pi_h u)'\|_{L^2}^2}_{\le C_1^2 h^2 \|u''\|^2} + \underbrace{\int_{-\pi}^{\pi} q \, (u - \pi_h u)^2 \, \mathrm{d}x}_{\le q_{\max} C_2^2 h^4 \|u''\|^2} \;\le\; h^2 \big(C_1^2 + q_{\max} C_2^2 h^2 \big) \|u''\|_{L^2}^2. \tag{29}$$

Combining (24) with $v = \pi_h u$ and (29),

$$\boxed{\|u - U\|_E \;\le\; \|u - \pi_h u\|_E \;\le\; C \, h \, \|u''\|_{L^2(-\pi,\pi)}, \qquad C := \sqrt{C_1^2 + 4\pi^2 q_{\max} C_2^2}} \tag{30}$$

so the CG(1) error in the energy norm is $O(h)$, i.e. **first order**. Here the residual h^2 left inside the square root in (29) was removed using $h = 2\pi/(M+1) \le 2\pi$, so that C depends only on $q_{\max}$ and the two interpolation constants — and not on the mesh, as an *a priori* constant should not.

Note. Two distinct ideas are doing the work, and it is worth separating them. (24) is about the *method*: Galerkin picks the best member of V_h, whatever V_h is. (28) is about the *space*: how well piecewise linears can represent u at all. The method contributes no error of its own — the entire $O(h)$ comes from the approximation power of the mesh. This is the rate to expect from the convergence table in Problem 5.

Problem 4

Problem. Prove an *a priori* error estimate in the L^2-norm.

Solution

Write $(v,w)=\int_{-\pi}^{\pi} v\,w\,\mathrm{d}x$ and $\|v\|=\sqrt{(v,v)}$ for the L^2 inner product and norm, and keep $e=u-U$. The energy estimate (30) is only $O(h)$. To gain a power of h we use a duality argument: instead of estimating e directly, we feed it as data into an auxiliary problem.

Step 1: the dual problem. Define ϕ by the variational problem

$$a(\phi,v)=(e,v)\qquad\forall\, v\in V, \tag{31}$$

which is (6) with the error e in place of the load f. The same bilinear form a appears on the left because the operator is self-adjoint — that is what makes the dual problem the *same* problem. Taking (31), rather than a strong form, as the definition keeps the logic in order. A unique $\phi\in V$ exists: a is bounded on V, coercive by (12) together with $q_{\min}>0$, and $v\mapsto(e,v)$ is a bounded linear functional, so Lax–Milgram applies. Regularity then *follows* instead of being assumed: testing (31) with $v\in C_c^\infty(-\pi,\pi)$ gives $\phi''=q\phi-e$ in the distributional sense, and the right-hand side lies in L^2, so $\phi\in H^2$. Only now is the strong form meaningful pointwise,

$$-\phi''+q(x)\,\phi=e\ \text{ a.e. on }(-\pi,\pi),\qquad \phi(-\pi)=\phi(\pi),\qquad \phi'(-\pi)=\phi'(\pi), \tag{32}$$

the periodicity of ϕ' being recovered from (31) exactly as BC2 was recovered in Problem 1. It is this H^2 regularity that Step 3 uses when it differentiates ϕ twice.

Step 2: the chain. Put $v=e$ in (31), use symmetry of a, then subtract $a(e,\pi_h\phi)=0$ — which is Galerkin orthogonality (11), legitimate because $\pi_h\phi\in V_h$ — and finish with the Cauchy–Schwarz inequality for the inner product a:

$$\begin{aligned}
\|e\|^2 &= (e,e)\\
&= a(\phi,e) && \text{by (31) with } v=e\\
&= a(e,\phi) && a \text{ is symmetric}\\
&= a\big(e,\ \phi-\pi_h\phi\big) && \text{since } a(e,\pi_h\phi)=0\\
&\le \|e\|_E\,\|\phi-\pi_h\phi\|_E && \text{Cauchy–Schwarz in } a.
\end{aligned} \tag{33}$$

By (29) applied to ϕ instead of u, $\|\phi-\pi_h\phi\|_E\le C\,h\,\|\phi''\|$, so

$$\|e\|^2\ \le\ C\,h\,\|e\|_E\,\|\phi''\|.$$

It remains to trade $\|\phi''\|$ for $\|e\|$.

Step 3: stability of the dual problem. First bound ϕ itself. Put $v=\phi$ in (31) and write $q_{\min}=\min_{[-\pi,\pi]} q>0$:

$$q_{\min}\|\phi\|^2\ \le\ \int_{-\pi}^{\pi}\Big[(\phi')^2+q\,\phi^2\Big]\,\mathrm{d}x\ =\ \|\phi\|_E^2\ =\ (e,\phi)\ \le\ \|e\|\,\|\phi\|,$$

so that

$$\|\phi\|\ \le\ \frac{1}{q_{\min}}\,\|e\|. \tag{34}$$

Now read ϕ'' off the differential equation (32) itself, $\phi'' = q\phi - e$, and use (34):

$$\|\phi''\| \;\le\; q_{\max}\|\phi\| + \|e\| \;\le\; \underbrace{\Big(1 + \frac{q_{\max}}{q_{\min}}\Big)}_{=:\,C_S}\|e\|. \tag{35}$$

Note. This is the step where the periodic problem genuinely differs from the Dirichlet one. In Lecture 8 the stability constant came from the **Poincaré inequality**, which bounds $\|\phi\|$ by $\|\phi'\|$. That inequality is *false* on the periodic space (5): a nonzero constant has $\phi' \equiv 0$ but $\|\phi\| \neq 0$. With no boundary condition pinning the function down, the zeroth-order term q is what supplies stability instead — which is why $q_{\min} > 0$ appears in (34). It is the same role $q > 0$ played in (12) and in the nonsingularity of $\boldsymbol{A}$.

Step 4: combine. Substituting (35) into the result of Step 2,

$$\|e\|^2 \;\le\; C\,C_S\,h\,\|e\|_E\,\|e\| \qquad\Longrightarrow\qquad \|e\| \;\le\; C\,C_S\,h\,\|e\|_E,$$

after cancelling one factor $\|e\|$ (if $\|e\| = 0$ there is nothing to prove). Finally insert the energy estimate (30):

$$\boxed{\|u - U\|_{L^2(-\pi,\pi)} \;\le\; \widetilde{C}\,h^2\,\|u''\|_{L^2(-\pi,\pi)}, \qquad \widetilde{C} = C^2 C_S} \tag{36}$$

so the CG(1) error in L^2 is $O(h^2)$, i.e. **second order** — one power of h better than the energy norm.

Note. Where the extra power came from: (33) pairs the error against the *interpolation error of the dual solution* rather than against itself, and interpolation costs one power of h. The duality step is what converts "e is small in energy" into "e is small in L^2". These are the two rates, $O(h)$ and $O(h^2)$, that the convergence table in Problem 5 has to reproduce.

Problem 5

Problem. Implement CG(1) numerically on a uniform mesh with

$$q(x) = 3 - \sin(x) - \sin(2x) - \cos(x), \quad f(x) = 2e^{\sin(x)}e^{\cos(x)}, \quad u(x) = e^{\sin(x)}e^{\cos(x)},$$

and determine the convergence rate in both the energy norm and the L^2-norm by a convergence table with $M = 24, 49, 99, 199, 399, 799$.

Solution

Code: `hw2_p5_cg1.py` (Python, `numpy`/`scipy`).

Verification of the data. Before solving anything the script checks that the stated u, q, f really are consistent. With $u = e^{\sin x + \cos x}$,

$$u' = u\,(\cos x - \sin x), \qquad u'' = u\big[(\cos x - \sin x)^2 - (\sin x + \cos x)\big],$$

and since $(\cos x - \sin x)^2 = 1 - \sin 2x$,

$$-u'' + q\,u = u\big[-1 + \sin 2x + \sin x + \cos x + 3 - \sin x - \sin 2x - \cos x\big] = 2u = f. \quad \checkmark$$

Numerically $\max|-u'' + qu - f| = 2.7 \times 10^{-15}$, and the periodicity defect $|u(-\pi) - u(\pi)| + |u'(-\pi) - u'(\pi)| = 3.3 \times 10^{-16}$, so u satisfies (1) including both boundary conditions.

Implementation. Following Problem 2: $N = M + 1$ unknowns and N elements, $h = 2\pi/N$. On the reference element $[0, 1]$ with $N_0(\xi) = 1 - \xi$, $N_1(\xi) = \xi$, the element contributions to (22) are

$$\boldsymbol{K}^e = \frac{1}{h}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \qquad M^e_{ab} = \int_{I_e} q\,N_a N_b\,\mathrm{d}x, \qquad b^e_a = \int_{I_e} f\,N_a\,\mathrm{d}x. \tag{37}$$

$\boldsymbol{K}^e$ is exact. Because q and f are not polynomials, M^e and b^e are evaluated with 6-point Gauss–Legendre quadrature per element, which is far more accurate than the $O(h^2)$ discretisation error and so does not pollute the observed rates.
The *only* periodic-specific line in the whole assembly is the local-to-global map:

$$\text{element } e \;\longmapsto\; \text{nodes } \big(e,\ (e+1) \bmod N\big), \qquad e = 0, 1, \ldots, N-1, \tag{38}$$

(the code indexes elements from 0, so element e here is the element called I_{e+1} in (14)), where the mod$N$ sends the last element's right node back to node 0. That single modulus is what produces the corner entries $A_{0,M} = A_{M,0}$ predicted in Problem 2 — no rows or columns are deleted, unlike the Dirichlet case.
The system is solved with `scipy.sparse.linalg.cg` (tolerance 10^{-13}), legitimate because $\boldsymbol{A}$ is symmetric positive definite by Problem 2. Both error norms are computed element by element with the same Gauss rule, using u' analytically and U' constant on each element.

Convergence table. The observed rate between two consecutive meshes is

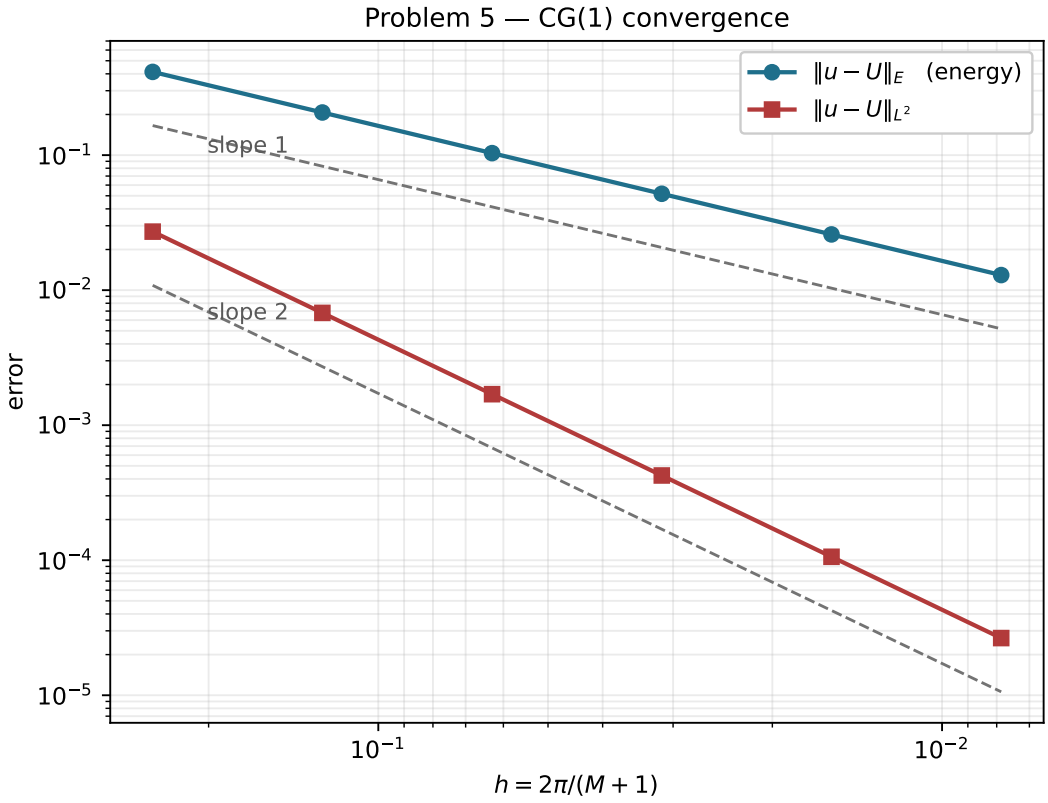
$$r_k = \frac{\ln\big(E_{k-1}/E_k\big)}{\ln\big(h_{k-1}/h_k\big)}. \tag{39}$$

M	h	$\|u-U\|_{L^2}$	rate	$\|u-U\|_E$	rate
24	2.5133e-01	2.7134e-02	—	4.1346e-01	—
49	1.2566e-01	6.7860e-03	2.00	2.0691e-01	1.00
99	6.2832e-02	1.6967e-03	2.00	1.0348e-01	1.00
199	3.1416e-02	4.2418e-04	2.00	5.1743e-02	1.00
399	1.5708e-02	1.0605e-04	2.00	2.5872e-02	1.00
799	7.8540e-03	2.6511e-05	2.00	1.2936e-02	1.00

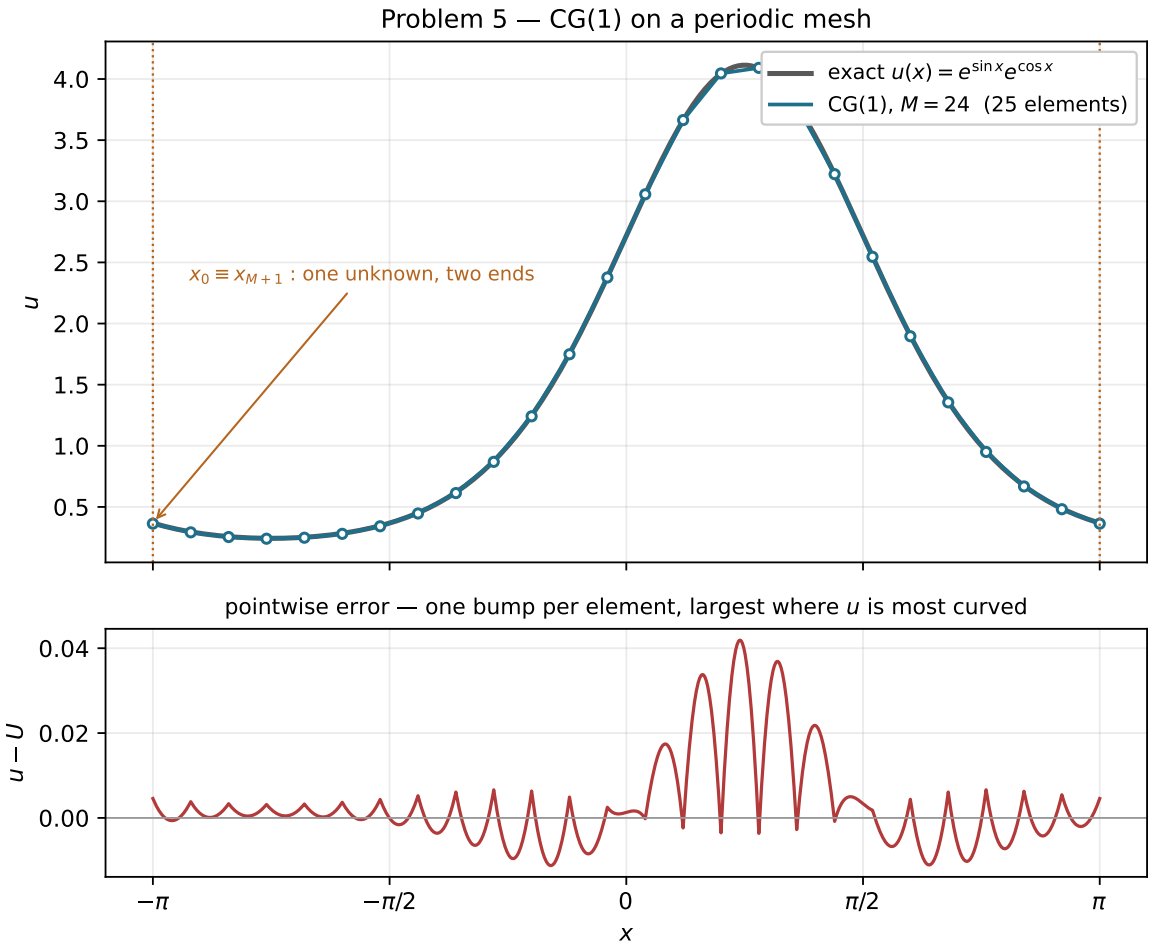
Conclusion. The observed rates are

$$\|u-U\|_E = O(h^{1.00}), \qquad \|u-U\|_{L^2} = O(h^{2.00}),$$

which is exactly the theory: first order in the energy norm from (30), and second order in L^2 from (36).



Both error curves are straight on a log–log plot and parallel to the reference slopes 1 and 2 (dashed, offset downward for legibility).



Top: exact solution and the CG(1) solution at $M = 24$; the circles are the 25 nodes. The dotted lines at $x = \pm\pi$ mark the *same* point of the domain — one unknown ξ_0 serving both ends, which is the wrap-around basis function (17) at work. **Bottom:** the pointwise error.

Note. The lower panel is where the structure shows: one bump per element, largest near $x \approx \pi/4$ where u is most strongly curved — the factor $\|u''\|$ of (30) visible directly.

It is worth *measuring* rather than assuming that the error vanishes at the nodes, because it does not. The script prints a nodal-error column: $\max_j |u(x_j) - U_j| = 6.6 \times 10^{-3}$ at $M = 24$, which is $0.245\,\|u - U\|_{L^2}$, and that ratio stays fixed under refinement — the nodal error decays at the same $O(h^2)$ rate as everything else, it is simply smaller. CG(1) *is* nodally exact for the pure problem $-u'' = f$, but the zeroth-order term $q\,u$ destroys that property; the mechanism is explained in the closing note of Problem 7, where the contrast is measured.

Problem 6

Problem. Implement the CG(2) method (piecewise quadratic polynomials) as discussed in class for

$$-u'' = f(x) \ \text{ on } (0,1), \qquad u(0) = u(1) = 0.$$

Solution

Code: `hw2_p67_cg2.py`.

Variational setting. The derivation of Problem 1 carries over with $q \equiv 0$ and periodicity replaced by Dirichlet conditions, and the optimality argument (24) of Problem 3 then applies verbatim, since it uses only Galerkin orthogonality and the fact that a is an inner product. Two things do *not* carry over: Problem 1's definiteness step used $q > 0$ and must be replaced (see the note below), and the *rate* in (30) was for piecewise linears, so it is the CG(2) estimates of Lecture 10 that apply here. The space is

$$V = \Big\{ v : \ \int_0^1 \big(v^2 + (v')^2\big)\,\mathrm{d}x < \infty, \ \ v(0) = v(1) = 0 \Big\},$$

and multiplying $-u'' = f$ by $v \in V$ and integrating by parts once kills the boundary term because v vanishes at *both* ends — here both conditions are essential, and there is no natural one. So: find $u \in V$ with

$$a(u,v) = \int_0^1 u'v'\,\mathrm{d}x = \int_0^1 f\,v\,\mathrm{d}x \qquad \forall\, v \in V, \qquad \text{hence} \qquad \|v\|_E^2 = \int_0^1 (v')^2\,\mathrm{d}x.$$

Taking $V_h \subset V$ and subtracting exactly as in (9) gives Galerkin orthogonality $a(u - U, v) = 0$ for all $v \in V_h$, so the optimality result (24) of Problem 3 applies verbatim here too.

Note. $\|\cdot\|_E$ really is a norm on this V even though there is no q term: $v' \equiv 0$ forces v constant, and $v(0) = 0$ then forces $v \equiv 0$. On $H^1(0,1)$ *without* the boundary conditions it would be only a seminorm. This is the Dirichlet counterpart of the remark closing Problem 1 — there it was $q > 0$ that eliminated the constants, here it is the boundary condition.

Mesh and numbering. With $h = 1/(M+1)$ there are $M+1$ elements and, as in Lecture 10, a submesh carrying one midside node per element. Index every node by a half-integer:

$$x_{k/2} = \tfrac{kh}{2}, \quad k = 0, 1, \ldots, 2M+2: \qquad k \text{ even } \Rightarrow \text{ vertex}, \qquad k \text{ odd } \Rightarrow \text{ midside}. \tag{40}$$

That is $2M+3$ nodes; the two Dirichlet ends $k = 0$ and $k = 2M+2$ are eliminated, leaving $2M+1$ **unknowns** — the size Lecture 10 predicts.

Basis. On the reference element $[0,1]$ the three Lagrange quadratics are

$$N_0(\xi) = 2\big(\xi - \tfrac{1}{2}\big)(\xi - 1), \qquad N_1(\xi) = 4\xi(1-\xi), \qquad N_2(\xi) = 2\xi\big(\xi - \tfrac{1}{2}\big), \tag{41}$$

taking the value 1 at $\xi = 0,\ \tfrac{1}{2},\ 1$ respectively and 0 at the other two. These are the three Lagrange polynomials of Lecture 10 written on a single element.

Element stiffness. With $\mathrm{d}N_a/\mathrm{d}x = h^{-1}\mathrm{d}N_a/\mathrm{d}\xi$,

$$K^e_{ab} = \frac{1}{h}\int_0^1 \frac{\mathrm{d}N_a}{\mathrm{d}\xi}\frac{\mathrm{d}N_b}{\mathrm{d}\xi}\,\mathrm{d}\xi, \qquad \frac{\mathrm{d}N_0}{\mathrm{d}\xi} = 4\xi - 3, \ \ \frac{\mathrm{d}N_1}{\mathrm{d}\xi} = 4 - 8\xi, \ \ \frac{\mathrm{d}N_2}{\mathrm{d}\xi} = 4\xi - 1,$$

which integrates exactly to

$$\boldsymbol{K}^e = \frac{1}{3h}\begin{bmatrix} 7 & -8 & 1 \\ -8 & 16 & -8 \\ 1 & -8 & 7 \end{bmatrix} \qquad \text{(ordering: vertex, midside, vertex).} \tag{42}$$

The load $b_a^e = \int_{I_e} f\, N_a \,\mathrm{d}x$ is evaluated with 8-point Gauss–Legendre quadrature, since f is not a polynomial.

Note. Lecture 10 instead replaces f by its interpolant $\widetilde{\Pi}_h f$, giving $f_i = \frac{h}{15}\big[f{+}8f{+}f\big]$ on the midside rows and $\frac{h}{30}\big[-f{+}2f{+}8f{+}2f{-}f\big]$ on the vertex rows. That choice does not change the *asymptotic* rates. Implementing it (see `interp_load_compare()` in the script) and differencing the two discrete solutions gives $\|U_{\text{interp}} - U_{\text{Gauss}}\| = O(h^4)$ in both norms: measured rates $3.97, 3.99, 4.00, 4.00, 4.00$ in L^2 and $4.01, 4.00, 4.00, 4.00, 4.00$ in the energy norm — two orders below the $O(h^2)$ energy error and one below the $O(h^3)$ L^2 error.

It is not, however, invisible in the table *as printed*, because the table starts before the two have separated. With the interpolated load the L^2 entry at $M = 24$ moves from 1.11×10^{-3} to 1.14×10^{-3}, and the first printed L^2 rate moves from 2.99 to 3.02. From $M = 99$ onwards the L^2 column agrees to three digits, and the energy column agrees to three digits at every M.

So there are two reasons to prefer accurate quadrature here, neither of them the asymptotic rate: it keeps the coarse rows clean, and it preserves the machine-exact vertex values discussed at the end of Problem 7 — with the interpolated load those degrade to $O(h^4)$ (3.2×10^{-4} at $M = 24$, rates 3.97–4.02; the 3.65 printed for the finest mesh is the roundoff floor, since $\mathrm{cond}(\boldsymbol{A}) = O(h^{-2})$ puts solver noise at $\sim 10^{-9}$ there).

Verification against the lecture. Assembling (42) over the elements and striking out the two Dirichlet rows and columns, the code prints $3h\boldsymbol{A}$ for $M = 4$ (a 9×9 system):

$$3h\,\boldsymbol{A} = \begin{bmatrix} 16 & -8 & & & & & & & \\ -8 & 14 & -8 & 1 & & & & & \\ & -8 & 16 & -8 & & & & & \\ & 1 & -8 & 14 & -8 & 1 & & & \\ & & & -8 & 16 & -8 & & & \\ & & & 1 & -8 & 14 & -8 & 1 & \\ & & & & & -8 & 16 & -8 & \\ & & & & & 1 & -8 & 14 & -8 \\ & & & & & & & -8 & 16 \end{bmatrix} \tag{43}$$

This reproduces Lecture 10 exactly: $16/3h$ on the midside diagonal, $14/3h$ on the vertex diagonal, $-8/3h$ on the first off-diagonal and $1/3h$ two places out from a vertex row. The matrix is symmetric positive definite, which is what `cg` requires. Its pattern is *pentadiagonal* rather than tridiagonal: raising the polynomial degree widens the band, because a vertex row now reaches the vertex two places away through the shared midside. That is a separate effect from the wrap-around corners of Problem 2, which came from periodicity and not from the degree — the CG(1) matrix there is not tridiagonal either, for a different reason.

Problem 7

Problem. Apply the method to

$$u(x) = e^{1-\cos(2\pi x)-\sin(2\pi x)} - 1, \qquad f(x) = 4\pi^2 e^{1-\cos(2\pi x)-\sin(2\pi x)} \Big[\sin(4\pi x) - \cos(2\pi x) - \sin(2\pi x) - 1\Big],$$

and determine the convergence rate in both norms for $M = 24, 49, 99, 199, 399, 799$.

Solution

Verification of the data. Write $g(x) = 1 - \cos(2\pi x) - \sin(2\pi x)$, so $u = e^g - 1$ and

$$g' = 2\pi\big(\sin 2\pi x - \cos 2\pi x\big), \qquad g'' = 4\pi^2\big(\cos 2\pi x + \sin 2\pi x\big), \qquad (g')^2 = 4\pi^2\big(1 - \sin 4\pi x\big).$$

Then $u'' = e^g\big[(g')^2 + g''\big]$, so

$$-u'' = 4\pi^2 e^g \Big[\sin 4\pi x - \cos 2\pi x - \sin 2\pi x - 1\Big] = f. \quad \checkmark \tag{44}$$

Also $u(0) = u(1) = e^0 - 1 = 0$, so both boundary conditions hold. Numerically $\max|-u'' - f| = 3.4 \times 10^{-13}$ and $|u(0)| + |u(1)| = 2.2 \times 10^{-16}$.

Convergence table.

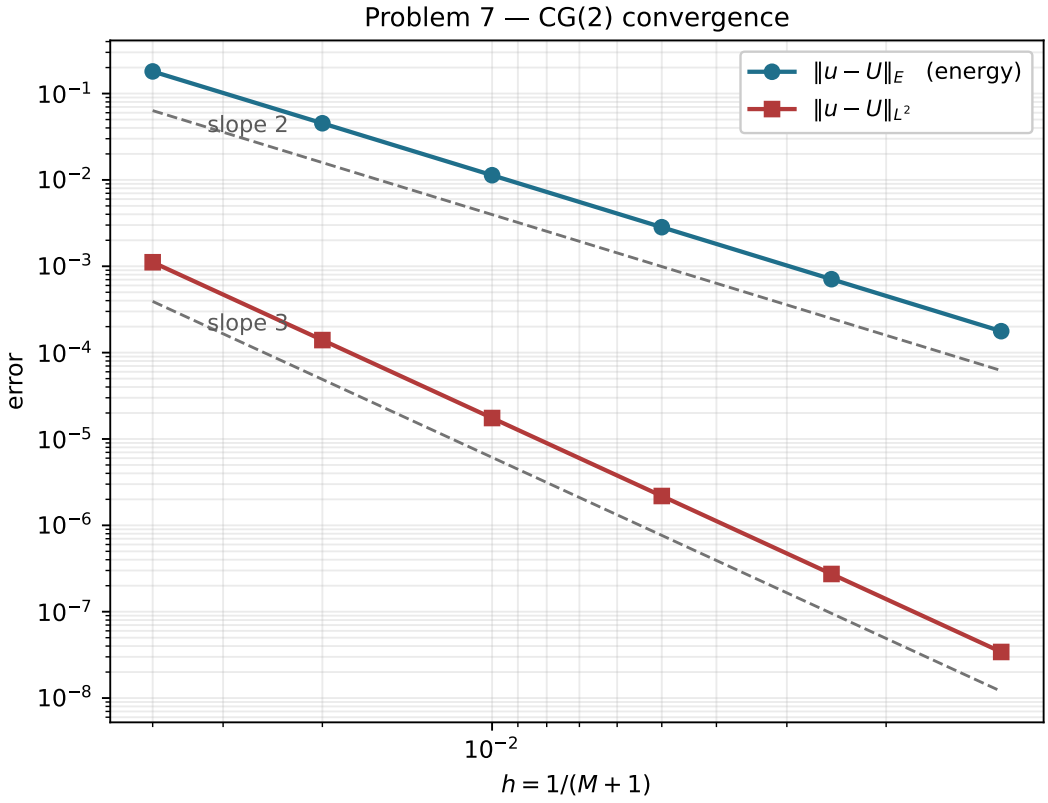
M	h	$\|u-U\|_{L^2}$	rate	$\|u-U\|_E$	rate
24	4.0000e-02	1.1132e-03	—	1.8041e-01	—
49	2.0000e-02	1.3975e-04	2.99	4.5288e-02	1.99
99	1.0000e-02	1.7488e-05	3.00	1.1334e-02	2.00
199	5.0000e-03	2.1866e-06	3.00	2.8341e-03	2.00
399	2.5000e-03	2.7334e-07	3.00	7.0857e-04	2.00
799	1.2500e-03	3.4168e-08	3.00	1.7715e-04	2.00

Conclusion. The observed rates are

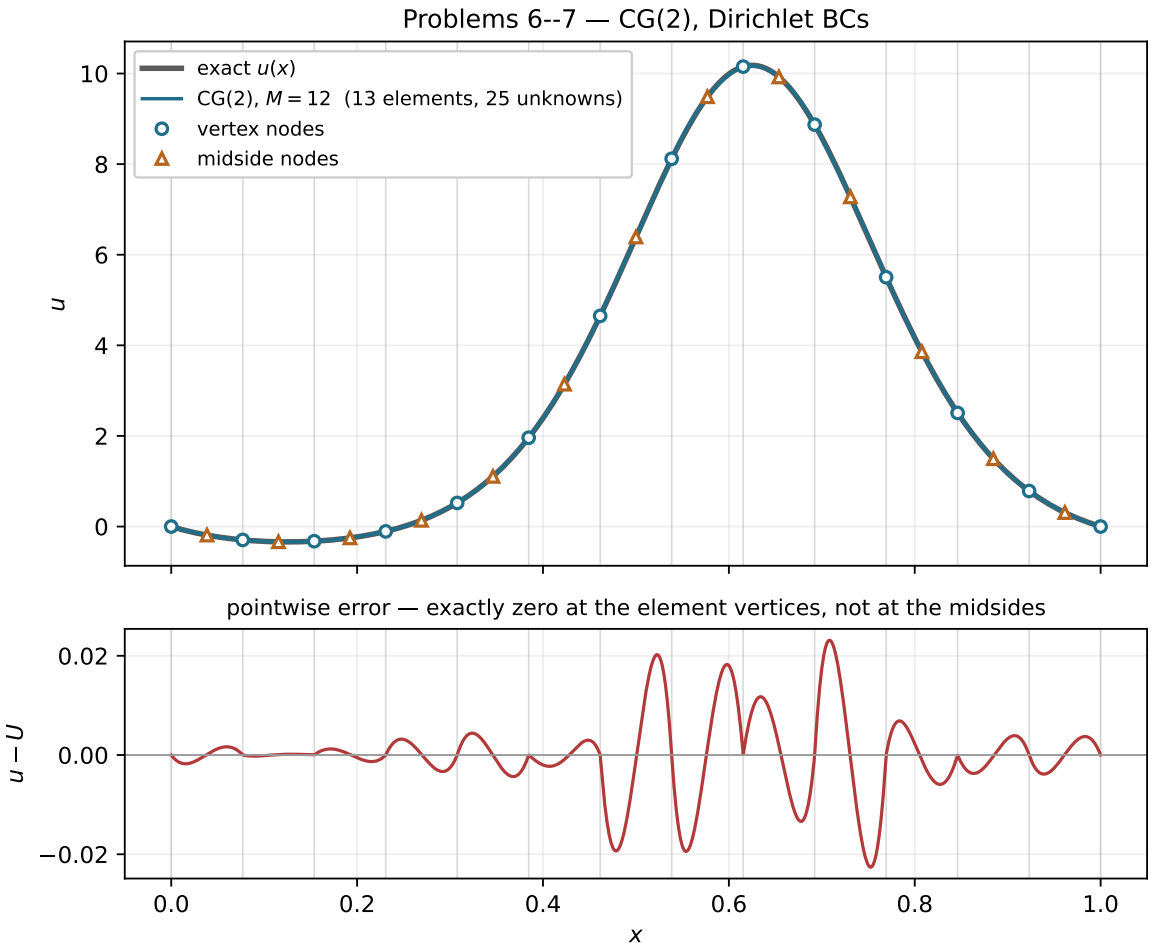
$$\|u-U\|_E = O(h^{2.00}), \qquad \|u-U\|_{L^2} = O(h^{3.00}),$$

matching the CG(2) estimates quoted in Lecture 10, $\|u-U\|_E \le c\|h^2 u'''\|$ and $\|u-U\| \le c\|h^3 u'''\|$. Each extra polynomial degree buys one power of h in both norms, and L^2 stays one order ahead of the energy norm. The *structure* of the duality argument of Problem 4 is what delivers that extra order, and it does not care what degree the elements are — but its stability step does not carry over unchanged: with $q \equiv 0$ the bound (34) is vacuous and $C_S = 1 + q_{\max}/q_{\min}$ is undefined. Here the dual problem is simply $-\phi'' = e$ with $\phi(0) = \phi(1) = 0$, so $\|\phi''\| = \|e\|$ is immediate and $C_S = 1$: the *stability* step needs neither $q_{\min}$ nor a Poincaré inequality.

Note. Poincaré does not disappear from the Dirichlet problem altogether, and it is worth saying where it goes. With $q \equiv 0$ it is what makes a coercive on V at all — $\int (v')^2 \ge c\|v\|_{H^1}^2$ requires it — so it takes over the role that $q_{\min} > 0$ played in the existence argument of Problem 4, Step 1. What it is *not* needed for here is the stability bound, because $-\phi'' = e$ hands over ϕ'' directly. For a variable coefficient, $-(a\phi')' = e$, that shortcut is unavailable and Poincaré returns to the stability step as well — which is the situation of Lecture 8 referred to in Problem 4.



Reference slopes 2 and 3 dashed, offset downward for legibility.



Top: exact and CG(2) solutions at $M = 12$; circles are vertex nodes, triangles midside nodes. **Bottom:** the pointwise error.

Note. Here the error *is* exactly zero at the element vertices: the script prints $\max_j |u(x_j) - U(x_j)| = 1.3 \times 10^{-13}$ at $M = 12$, i.e. roundoff — but *not* at the midsides, where it is 1.9×10^{-3}.
Why. It is the Green's function. Let $G(\,\cdot\,; x_j)$ solve $-G'' = \delta(\cdot - x_j)$ with homogeneous Dirichlet ends — the function computed in HW1, Problem 1(d). It is piecewise linear with its only kink at x_j, and it satisfies $a(v, G(\,\cdot\,; x_j)) = v(x_j)$ for every $v \in V$. If x_j is a mesh *vertex*, then $G(\,\cdot\,; x_j)$ is linear on every element, hence $G(\,\cdot\,; x_j) \in V_h$, and therefore

$$u(x_j) - U(x_j) = a\big(u - U,\ G(\,\cdot\,; x_j)\big) = 0$$

by Galerkin orthogonality. Three hypotheses are visible in that one line, and each one fails somewhere in this homework:

- the operator must be $-\mathrm{d}^2/\mathrm{d}x^2$. With a zeroth-order term the Green's function of $-G'' + qG = \delta$ is no longer piecewise linear, so it leaves V_h — which is why Problem 5 has no such exactness;
- the point must be a mesh *vertex*. If x_j is a midside then the kink of $G(\,\cdot\,; x_j)$ falls in the *interior* of an element rather than at a mesh breakpoint, so G is not a polynomial on that element and leaves V_h — hence the 1.9×10^{-3};
- the load must be integrated exactly, or U is not the Galerkin solution and the orthogonality fails. With Lecture 10's interpolated load the vertex error degrades to $O(h^4)$, as noted in Problem 6.

So vertex exactness is a joint property of the operator, the mesh and the quadrature — not something the finite element method supplies on its own. It is also one-dimensional: the argument works only because the

Green's function of a point load is itself a continuous piecewise polynomial. In two dimensions the Green's function of $-\Delta$ is log-singular and not even in H^1, so it is never in V_h and there is no nodal exactness to be had.